# COFRAME RIGIDITY AND FINITE STATIONARY SEPARATION FOR SKEW SCHUR FUNCTIONS

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Abstract. The skew Schur function of a connected diagram, together with the function obtained by splicing the same fixed connected cap to it, determines the diagram up to translation. We prove this using complete rational Jacobi–Trudi coframes, retaining every unit and inverse pole. For zero-free diagrams and caps, a separate linearized theorem detects every nonzero constant-weight combination of vacancy restorations from the two observations. These results imply finite stationary separation: a shifted positive-power identity with a fixed rational potential forces two equal-function connected diagrams to coincide. Its precision is fixed by the supplied span and includes the terminal coefficient. We also treat a whole moving potential by field descent and prove an unshifted exclusion for opposite ends of one diagram. A weak-cap theorem recovers absolute flags, including units, from the complete rational coframe.

## 1. Introduction

A skew Schur function need not determine a connected skew diagram up to translation or rotation [2]. Even rotation can hide boundary data needed for reconstruction: the two hooks $(2,1)$ and $(2,2)/(1)$ have the same function but different full beta lists. The question here is which complete rational boundary observations restore that information.

A Jacobi–Trudi coframe records the entire solution of a boundary-column system. Its support remembers the native columns, and its product with the complete-generator series remembers the native rows. Entries of index zero belong to both lists and remain in the matrix. Deleting them before solving changes the rational observation, even when a leading homogeneous determinant is unchanged.

For connected diagrams $B, K$, the 00-splice $K \triangleleft B$ joins their full boundary lists at a position occupied by neither internal word. Its precise lists are (3.1). Our fixed-cap theorem, Theorem 3.2, says that

$$B \longmapsto (s_B, s_{K \triangleleft B})$$

is injective on connected normalized diagrams for each fixed literal connected $K$. The same cap must be used on both sides; equality of cap functions is not the hypothesis. Figure 1 illustrates the operation on the equal hook pair.

Nonlinear injectivity does not imply injectivity of a linearization. For a zero-free diagram, a vacancy is an internal beta position absent from both lists. Restoring its row and column produces a bordered determinant. Theorem 4.3 proves independently that the whole child and same-cap variations detect every nonzero combination of these restorations with constant rational weights, provided both child and cap are zero-free. The rational weights allowed in the native-equation calculation of Lemma 4.1 are a different contract.

The finite stationary theorem uses the clipped alternating operator $\mathcal{T}_{G,C}$ defined in (2.2). For complete coframes of actual connected diagrams with equal whole functions, Theorem 5.4 proves that

$$\mathcal{T}_{G,C}(zH^p) + z\mathcal{T}_{G,C}(H)r(H) = 0 \pmod{z^{2M}}, \qquad p \geq 2,$$

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forces literal equality. Here $M$ is the supplied span and $r$ is a fixed rational function over a field independent of the positive raw letters. The clock is transcendental over that field. The coefficient of degree $2M-1$ is part of the hypothesis; the equal hooks in Example 5.6 show how the potential can first occur there. Corollary 5.5 allows a potential in a larger neutral field when a whole moving homomorphism is supplied. Every dependent coordinate and inverse entry moves under that map.

The opposite-end equation in Lemma 6.1 is unshifted and concerns one diagram and its rotation. It is not the preceding theorem with the initial $z$ canceled. Finite clipping contributes the whole cap polynomial in (2.3). Its proof retains connected and disconnected proper corners, genuine units, and the exponent-two endpoint exceptions.

Connected skew-Schur irreducibility is used in its corrected form due to Barekat, Reiner, and van Willigenburg [1]; overlap and shape invariants are from [2]. The two-boundary reconstruction results of [3] use first-row deletion and overlap-one row extension. The present 00-splice instead replaces the top beta when the cap is a row, and is not that extension.

Section 2 establishes the full-flag coordinates, coframe atlas, defect bounds, and weak-cap reconstruction. Section 3 proves fixed-cap rigidity. Section 4 treats complete unit restoration, its independent linearized theorem, and the first cell-splice grade. Section 5 matches highest components and proves shifted separation and whole-field descent. Section 6 proves the opposite-end exclusion.

## 2. Complete coframe coordinates

2.1. **Diagrams, full flags, and coherent units.** A diagram is a finite skew Young diagram in English notation, considered up to translation. Connected means nonempty and edge-connected. For $A=\lambda/\mu$ with $r$ nonempty rows, its full beta lists are $X_i=\lambda_i+r-i$ and $Y_i=\mu_i+r-i$, with a common translation chosen so that $Y_r=0$. Both lists are strictly decreasing. The span $M=X_1$ is the number of occupied content diagonals. The physical function is

$$s_A=\det(h_{X_i-Y_j})_{i,j=1}^{r},\qquad h_0=1,\quad h_j=0\ (j<0),$$

over $\mathbb{Q}[h_1,h_2,\dots]$. The full flags satisfy $X_i>Y_i$ and $X_{i+1}\geq Y_i$; these are the nonempty-row and adjacent-row connectivity inequalities. Equal labels in different lists are retained. The boundary height at $0\leq k<M$ is $\#\{y\in Y: y\leq k\}-\#\{x\in X: x\leq k\}>0$. A diagram is *zero-free* when $X\cap Y=\varnothing$. A *genuine vacancy* is a position in $\{1,\dots,M-1\}$ absent from both full lists. The notation $A+b$ means insertion of $b$ into both lists, followed by the actual separate sorting. Write $A^*$ for its normalized half-turn; its lists are $M-Y, M-X$, respectively. Also write CELL for the one-cell diagram and $\mathrm{Row}_j$ for the row of length $j$.

The proper matrix deletes the top row beta $M$ and the column beta zero from the complete matrix. If the full rank is $r$, its *signed proper corner* is $\kappa_A=(-1)^{r-1}\det T_A$. The determinant of the empty proper matrix is one. A weak proper matrix may split into connected positive-span blocks and one-by-one blocks consisting of the true unit. We call these its strict components, retaining their absolute row and column positions and all crossing entries of the full triangular matrix. A unit has span zero; a cell has span one and determinant $h_1$, not one.

Coherent normalization replaces every zero-index entry by the same coordinate $\vartheta$, not by separate variables. The notation $s_A$ in a coherent alphabet denotes the resulting determinant; physical normalization means $\vartheta=1$. The same conventions apply to all proper blocks. The absolute adjugate entry keyed by $(x_i,y_j)$ is the signed cofactor $(-1)^{i+j}$ times the minor with row $i$ and column $j$ deleted, and is zero at absent flags. Thus differences of absolute arrays compare equal beta positions, not matrix indices after a deletion.

For a series $u$, write $[u]_j=[z^j]u$, and let $\operatorname{ord}_z u$ be its least nonzero exponent. All coefficient fields below have characteristic zero. Positive raw letters are algebraically independent; when a coherent top letter is needed it too is independent of the proper raw field. A top-neutralization means solving the affine top equation $e=s_A=0$ for $H_M$, with $\kappa_A=\partial_{H_M}e\neq 0$. No proper inverse denominator is set to zero by this substitution.

2.2. **The complete boundary solve.** Let $A$ be a connected normalized skew diagram of span $M$. Write its full beta lists as

$$X_A = \{M\} \cup I_A, \qquad Y_A = S_A \cup \{0\}, \qquad I_A, S_A \subseteq \{1, \dots, M-1\}.$$

Thus $I_A$ and $S_A$ have the same cardinality. Throughout this section, $K$ is a field of characteristic zero, $\vartheta$ is transcendental over $K$, and the positive letters $H_1, \dots, H_{M-1}$ are algebraically independent over $K(\vartheta)$. Set $H_0 = \vartheta$ and $H_j = 0$ for $j < 0$. In particular, every actual zero-index entry of a Jacobi–Trudi matrix is the same coherent coordinate $\vartheta$. It is not replaced by an independently variable unit.

The *complete right coframe* of $A$ is

$$G(z) = 1 + \sum_{s \in S_A} G_s z^s, \qquad (G_s)_{s \in S_A} = -(H_{i-s})^{-1}_{i \in I_A, s \in S_A} (H_i)_{i \in I_A}. \tag{2.1}$$

The empty matrix gives $G = 1$. All inverses in this definition are taken in the full rational function field. In particular, their poles, their genuine units, and the contributions of every crossing remain part of the coframe. Let $C$ be the corresponding coframe of a second diagram $B$ of the same span. For a series $u$, write

$$\mathcal{T}_{G,C}(u) = G[uC]_{<M} - C[uG]_{<M}, \qquad D = \mathcal{T}_{G,C}(H), \qquad H = \vartheta + \sum_{i=1}^{M-1} H_i z^i + O(z^M). \tag{2.2}$$

Here $[u]_{<M}$ denotes truncation to degrees strictly below $M$. The cutoff $M$ is fixed. The stationary equation is observed modulo $z^{2M}$, including its coefficient of degree $2M-1$.

The complete left coframe is the right coframe of the normalized rotation in the same alphabet. Equivalently, its coefficient at $M-i$, $i \in I_A$, is the $i$-entry of $-(H_{M-s})_{s \in S_A} T_A^{-1}$, and its constant term is one.

For arbitrary complete coframes and any series $u$, finite clipping also gives the exact polynomial identity

$$\mathcal{T}(zu) - z\mathcal{T}(u) = -z^M \{G[uC]_{M-1} - C[uG]_{M-1}\}. \tag{2.3}$$

Indeed, $[zuC]_{<M} = z[uC]_{<M} - z^M [uC]_{M-1}$, and the same identity holds with $G$ in place of $C$. Subtracting proves the formula. The condition is the zero whole cap polynomial, not separate vanishing of its two scalar coefficients.

2.3. **Row degree, proper corners, and independent coordinates.** For a coherent Jacobi–Trudi polynomial, give each positive letter $H_j$ weight $j$, and give $\vartheta = H_0$ weight zero. Its *positive row degree* is its maximum ordinary degree in the positive letters, with $\vartheta$ treated as a coefficient.

**Lemma 2.1.** *The positive row degree of any nonempty skew diagram is its number of nonempty rows. If the diagram is connected, its largest generator is its span $M$, and the coefficient of $H_M$ is its nonzero signed proper corner. A connected skew Schur polynomial is primitive; its coherent homogenization is irreducible over $\mathbb{Q}(\vartheta)$ and is relatively prime to its proper corner.*

*Proof.* For the row-degree assertion, factor over connected components. Their nonempty row counts and positive row degrees add, and coherent homogenization does not change that degree. It therefore suffices to consider a connected diagram. Let $X = (x_1 > \cdots > x_r)$ and $Y = (y_1 > \cdots > y_r)$ be the full lists. For a permutation $\pi$, the square weight of its determinant term is

$$\sum_i (x_i - y_{\pi(i)})^2 = \sum_i x_i^2 + \sum_i y_i^2 - 2\sum_i x_i y_{\pi(i)}.$$

Strict rearrangement makes the identity permutation the unique minimizer. The diagonal indices are positive, so its monomial survives with coefficient one and ordinary degree $r$. No term has larger degree. Factorization then proves the general row-degree assertion. For a connected diagram the complementary extreme-corner matrix has diagonal indices $x_{i+1} - y_i \geq 0$, by adjacent-row connectivity. Strict rearrangement on these remaining lists again gives a unique minimum-square monomial, now allowing the coherent unit at zero, and proves that this corner

is nonzero. The unique largest matrix index $M = x_1 - y_r$ is then the largest generator of the determinant.

The surviving coefficient-one monomial proves primitivity. We use connected skew-Schur irreducibility in its corrected form [1]. The coherent polynomial is the homogenization

$$\vartheta^r s_A(H_1/\vartheta, H_2/\vartheta, \ldots).$$

It has no factor $\vartheta$, since its degree-$r$ part is nonzero. Homogenization therefore preserves irreducibility. Its proper corner is independent of $H_M$ and is nonzero, so it is relatively prime to the whole connected polynomial. □

In particular, equality of physical skew Schur functions determines the row count, and hence implies equality of their coherent homogenizations. Primitive normalization and evaluation at a sufficiently large positive alphabet identify any scalar occurring between two proportional actual connected Schur polynomials.

**Lemma 2.2.** *For the complete coframe $G$ in (2.1), set $P = [HG]_{<M}$. Then*

$$G_s \neq 0 \iff s \in S_A, \qquad P_i = 0 \iff i \in I_A \quad (0 < i < M). \tag{2.4}$$

*The native coefficients $G_s$, $s \in S_A$, together with the available numerator coefficients $P_i$, $0 < i < M$ and $i \notin I_A$, are algebraically independent coordinates over $K(\vartheta)$. Their rational inverse is*

$$H = P/G \pmod{z^M}, \qquad P_0 = \vartheta. \tag{2.5}$$

*If $\kappa_A$ is the signed proper corner, the whole coherent function satisfies $s_A = \kappa_A[HG]_M$, where the independent top letter $H_M$ is included in this last expression.*

*Proof.* The proper matrix has a nonzero determinant by Lemma 2.1; its weak diagonal includes any genuine units. The native equations give $P_i = 0$ for $i \in I_A$. Conversely, convolution and $G_0 = 1$ recover the raw coefficients successively by (2.5). Thus the listed coordinates generate the full field $K(\vartheta, H_1, \ldots, H_{M-1})$. Their number is exactly $M - 1$, its transcendence degree. They are therefore algebraically independent. In particular, every listed native or available coefficient is nonzero as a rational function. The last identity is the Schur complement at the unique top letter, with its actual Laplace sign. □

**Lemma 2.3.** *A nonconstant complete native coframe is irreducible over its full raw field. Distinct normalized coframes of actual coequal connected diagrams are coprime, and their defect $D$ is nonzero. Equal complete coframes identify the two diagrams literally.*

*Proof.* In the independent atlas a nonconstant coframe has the form $1 + \sum_{s \in S} u_s z^s$. Regard it first as a polynomial linear in one native variable $u_c$. Its two coefficients, $z^c$ and $1 + \sum_{s \neq c} u_s z^s$, are coprime. Gauss's lemma, first in $u_c$ and then in $z$, proves irreducibility over the field of the native coordinates. Adjoining the independent numerator coordinates does not change it. Distinct coframes with constant term one are therefore coprime. Equal coframes give equal $S$ by support and equal $I$ by the numerator-zero test in (2.4).

A constant coframe means $S = I = \varnothing$, so the diagram is a row. An actual coequal mate has the same row count and span and is the same row. Thus a distinct coequal pair has nonconstant coframes. In its independent atlas, $G$ is coprime to $P = [HG]_{<M}$: specialize the available positive numerator coordinates to zero, leaving $P = \vartheta$. Equivalently, the corresponding resultant has a nonzero specialization. If $D = GQ - CP = 0$ for a distinct pair, coprimality of $G, C$ would force $G \mid P$, a contradiction. At spans one and two the connected diagrams are the cell, the two-cell row and the two-cell column, with distinct functions where applicable. Thus a distinct coequal pair has $M \geq 3$. □

**Lemma 2.4.** *Suppose $s_A = s_B$. Their coherent Jacobi–Trudi polynomials have the same row degree and the same signed proper corner. Their monic top quotients agree, and*

$$\operatorname{ord}_z D \geq M + 1, \qquad \deg_z D \leq 2M - 2. \tag{2.6}$$

*If $A \neq B$ literally, then $M \geq 3$, $G$ and $C$ are distinct nonconstant coprime polynomials, and $D \neq 0$.*

*Proof.* The positive row degree, independent coframe coordinates, and top-quotient identities are Lemmas 2.1 to 2.3. The two coframes have constant term one. Consequently their two products with $H$ agree with their truncations below degree $M$. Equality of their degree-$M$ top quotients cancels that degree as well, which proves the order bound. The degree bound follows directly from (2.2). The last assertions are Lemma 2.3. □

The higher coefficients of $H$ may be retained from the whole returned series. They do not affect $D$ or either clipped term in $\mathcal{T}(zH^p)$, for any integer $p \geq 2$. For a fixed rational $r(x) \in K(x)$, (2.6) shows that changing them alters $zDr(H)$ only in degrees at least $2M+2$. Thus the finite equation below depends only on the displayed coefficients, while the whole moving map may keep all its higher jets.

2.4. **Weak caps and absolute flags.** Here $h_0 = 1$. The lists in the following result need only be weakly feasible: their diagonal differences may be zero and their proper blocks need not be connected. The numerator irreducibility asserted here is in the joint polynomial ring, not a consequence of specializing a coherent irreducible polynomial.

**Lemma 2.5** (Weak-cap recognition and absolute flags)**.** *Let $X_1 > \cdots > X_r > 0$, $Y_1 > \cdots > Y_r > 0$, $X_i \geq Y_i$, and put*

$$A = (h_{X_i - Y_j}), \quad C = \det A, \quad b = (h_{X_i}), \quad N_A(z) = C - (z^{Y_j}) \operatorname{adj}(A) b, \quad g_A = N_A / C.$$

*For a nonempty cap, $N_A$ is irreducible in the joint polynomial ring in $z, h_1, h_2, \ldots$. The whole rational coframe $g_A$ determines all the absolute flags $X, Y$, including units. The empty cap has $g_A = 1$.*

*Proof.* Let $a = X_1$. It is a fresh border label, occurring only in $b_1$, since all $Y_j$ are positive. Write $N_A = -h_a D + R$, with $D = (z^{Y_j}) \operatorname{adj}(A) e_1 \neq 0$. A common factor independent of $z$ would divide $C$ and every entry of $\operatorname{adj}(A)b$. For $T > X_1 + Y_1$, the lists $(T, X)/(Y, 0)$ define a connected actual board with determinant

$$h_T C - \sum_j h_{T - Y_j} (\operatorname{adj}(A) b)_j.$$

The first-row labels are fresh and distinct. Such a common factor would divide this connected Schur polynomial, contradicting connected irreducibility [1].

Suppose an irreducible positive-$z$-degree factor $F$ divides $D, R$. It is not $z$, since $N_A(0) = C \neq 0$. Work in the fraction field of the quotient by $F$; every nonzero $z$-free minor remains nonzero there. If $r = 1$, then $D = z^{Y_1}$ and $R = C$ is $z$-free, already a contradiction. Assume $r \geq 2$.

At stage $t$, $2 \leq t \leq r$, the invariant is

$$(z^{Y_j})_{j=1}^r = \sum_{i=t}^r \alpha_i A_{i,*}, \qquad \sum_{i=t}^r \alpha_i h_{X_i} = 1.$$

The first stage follows from $((z^{Y_j})A^{-1})_1 = 0$ and $(z^{Y_j})A^{-1}b = 1$, which are the equations $D = R = 0$. The trailing minor $B_t = A[t, \ldots, r; t, \ldots, r]$ is weakly feasible and nonzero. It determines the coefficients by

$$(\alpha_t, \ldots, \alpha_r) = (z^{Y_j})_{j=t}^r B_t^{-1}.$$

Its entries use only raw labels below $X_t$. All minors obtained by adjoining $(z^{Y_j})_{j=1}^r$ to the tail rows vanish modulo $F$. At least one is a nonzero polynomial: otherwise a nonzero $z$-free annihilator of that proper tail row space would annihilate the distinct powers $z^{Y_j}$ identically. Choose such a minor. It uses only labels below $X_t$, so its irreducible divisor $F$ is independent of $h_{X_t}$. The representation coefficients have the same independence. Taking the coefficient of the fresh $h_{X_t}$ in the second invariant therefore gives $\alpha_t = 0$ modulo $F$, advancing the invariant to $t+1$. After the last row is removed, every $z^{Y_j}$ is zero modulo $F$. This forces $F = z$, a contradiction. Hence $\gcd(D, R) = 1$, and Gauss's lemma proves irreducibility of $N_A$.

For reconstruction put $v = \operatorname{adj}(A)b$. Every $v_j$ is nonzero, since replacing column $Y_j$ by 0 and sorting gives a weakly feasible matrix. The positive $z$-support of $N_A$ recovers $Y$ exactly. Its largest raw label is $X_1$, and

$$D_j = [z^{Y_j}](-\partial_{h_{X_1}} N_A) = (\operatorname{adj}(A)e_1)_j.$$

For $T = A[2, \dots, r; 2, \dots, r]$, $D_1 = \det T \neq 0$, and the block equations give

$$(\operatorname{adj}(T)b_{\text{tail}})_{j-1} = \frac{D_1 v_j - D_j v_1}{C}, \qquad j \geq 2.$$

This reconstructs the entire tail numerator and proceeds recursively to the empty cap. Finally $g_A$ is reduced, since its numerator is irreducible and depends on a fresh border label absent from $C$. Integral primitivity and the normalization $g_A(0) = 1$ fix the associate. This proves the claim. □

## 3. Rigidity with the same literal cap

For this section use physical letters $h_0 = 1$ and independent $h_j$ for $j > 0$. All statements are identities in the polynomial or rational function fields of these letters. For connected normalized diagrams $B, K$ of spans $c, a$, respectively, define the actual 00-*splice* $K \triangleleft B$ by

$$\begin{aligned} X_{K\triangleleft B} &= (c + X_K) \cup (X_B \setminus \{c\}), \\ Y_{K\triangleleft B} &= (c + (Y_K \setminus \{0\})) \cup Y_B. \end{aligned} \tag{3.1}$$

The row lengths and adjacent connected inequalities of these full lists follow from those of $K$ and $B$. Thus this is an actual connected skew diagram of span $c + a$ and area $|K| + |B|$. It is an auxiliary splice; it is not being declared a composition chart.

Let $G_B$ be the complete right coframe of $B$. Let $C_K$ be the complete left coframe of $K$: if $T_K$ is its proper matrix, its coefficients at exponents $a - u$, for $u \in X_K \setminus \{a\}$, are the entries of $-(h_{a-v})_{v \in Y_K \setminus \{0\}} T_K^{-1}$, and its constant is one. A rectangular block expansion gives

$$s_{K\triangleleft B} = \kappa_K \kappa_B [z^{c+a}] \mathsf{H}(z) C_K(z) G_B(z), \qquad \mathsf{H}(z) = \sum_{j \geq 0} h_j z^j. \tag{3.2}$$

All matrices here are complete. In particular, (3.2) holds with units and negative-index zeros.

**Example 3.1.** Take $K = \mathrm{CELL}$, with lists $X_K = (1)$, $Y_K = (0)$. For $A = (2, 1)$ and $B = (2, 2)/(1)$ the full lists are $(3, 1)/(1, 0)$ and $(3, 2)/(2, 0)$. Formula (3.1) gives

$$X_{K\triangleleft A} = (4, 1), \quad Y_{K\triangleleft A} = (1, 0), \qquad X_{K\triangleleft B} = (4, 2), \quad Y_{K\triangleleft B} = (2, 0).$$

The splices are $(3, 1)$ and $(3, 2)/(1)$, with determinants

$$s_{K\triangleleft A} = h_1 h_3 - h_4, \qquad s_{K\triangleleft B} = h_2^2 - h_4.$$

Thus the same literal cell separates this equal-function pair. The common beta in each board is retained, at one and two respectively. Figure 1 shows the two literal splices.

**Theorem 3.2** (Fixed-cap rigidity)**.** *Let $B, B'$ be connected normalized skew diagrams and $K$ the same fixed literal connected cap. If*

$$s_B = s_{B'}, \qquad s_{K\triangleleft B} = s_{K\triangleleft B'},$$

*then $B = B'$ literally.*

*Proof.* We use strong induction on the child span $c$, uniformly over every actual connected cap. The common whole function supplies the common span, row count, and signed corner by Lemma 2.1. Put $Q = G_B - G_{B'}$. The two whole identities become

$$[z^c] \mathsf{H} Q = 0, \qquad [z^{c+a}] \mathsf{H} C_K Q = 0. \tag{3.3}$$

Every coefficient of $Q$ uses raw letters of index below $c$. If its current raw ceiling is $R$, put $R' = \max(R, a - 1)$. Taking the independent coefficients of $h_{R'+1}, \dots, h_{c+a}$ in the second equation, and removing $C_K$ successively from its constant term one, gives

$$Q_j = 0 \qquad (j < c + a - R'). \tag{3.4}$$

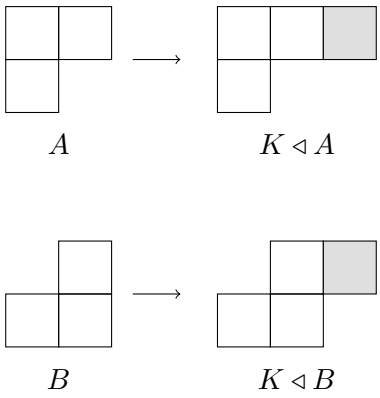



FIGURE 1. The same cell cap in Example 3.1. Shaded cells are added by the literal 00-splice. The top beta changes from three to four; no new row is introduced. The common row-column beta remains at one in the upper pair and at two in the lower pair.

If $R < a$, this already gives $Q = 0$.

We next identify the proper-corner components in decreasing order of their actual positions. Suppose all higher components have already been identified literally. Let $P$ be their entire common proper matrix, including its cross entries, with largest row $m_0$, least column $b$, and span $S = m_0 - b$. For an empty prefix omit the corresponding rows and equations. Let $t$ be the largest remaining proper row on either side. The lower equations are triangular with respect to the prefix, so their full coefficient difference uses raw letters only through $t$. Subtraction of the complete prefix equations gives

$$Q_{\mathrm{prefix}} = -P^{-1} \left( \sum_{j<b} h_{x-j} Q_j \right)_{x \in \mathrm{rows}\, P}. \tag{3.5}$$

If $L > \max(S, t)$, neither $P$ nor the lower coefficients depends on $h_L$. Differentiating a cross entry can only expose $Q_{x-L}$, which is zero by (3.4), because $x - L \le m_0 - L < c + a - L$. Descending the raw ceiling in the entire equation therefore earns $R = \max(S, t)$. If $S \ge t$, the zero range reaches beyond $b$, killing the lower part and then the prefix. Otherwise $R = t \ge a$.

At $h_t$, all remaining proper matrix entries have index strictly below $t$. The differentiated crossing entries in (3.5) again multiply coefficients already in the zero range. The derivative of the first equation in (3.3) has no direct term, since $Q_{c-t} = 0$. It consequently compares the two complete inverse columns selected by the next actual row $t$. If this row were absent on one side, the other would be a nonzero feasible Jacobi–Trudi coface, which is impossible.

Let $E, E'$ be the next strict proper blocks at $t$. A block with function one is its actual unit. In the full triangular inverse column every lower suffix determinant cancels as a whole. The remaining connected numerator has flags

$$X_N = \{c\} \cup \mathrm{rows}\, P \cup (\mathrm{rows}\, E \setminus \{t\}), \qquad Y_N = \mathrm{cols}\, P \cup \mathrm{cols}\, E. \tag{3.6}$$

Its positivity and connectedness follow by retaining the original prefix interval and the strict component interval. Its span is larger than that of every denominator component. Unique factorization and connected primeness therefore give equal functions of $E, E'$ and equal functions of their whole connected numerators. For a unit this identifies the unit literally. Otherwise equal span gives their common least column $b_0$.

The lists

$$X_{K_E} = \{c - t\} \cup (\mathrm{rows}\, P - t), \qquad Y_{K_E} = (\mathrm{cols}\, P - t) \cup \{0\} \tag{3.7}$$

form an actual connected cap. After translation by $b_0$, the numerators in (3.6) are literally $K_E \triangleleft E$ and $K_E \triangleleft E'$. The child span is $t - b_0 < c$. The induction hypothesis identifies the two blocks literally, and the common prefix can be extended. Every extension removes a proper

block; the process terminates. Thus $Q = 0$. Its support identifies all columns, and the numerator test in (2.4) identifies all proper rows. The common top completes the literal equality of $B$ and $B'$. □

## 4. Complete unit restoration

4.1. **The complete unit expansion and finite injectivity.** Deleting a common full row and column beta preserves the boundary height word. Deleting all common betas therefore gives a connected zero-free diagram of the same span and area, called the zero-free core. The deleted positions are actual vacancies of that core.

**Lemma 4.1.** *Scale every positive raw letter by $\epsilon$ while keeping every genuine unit coherent. If $A$ has row rank $r$ and $u$ common row-column positions, then*

$$s_A = \sigma_A \epsilon^{r-u}(s_{A_0} + O(\epsilon)) \tag{4.1}$$

*at $\vartheta = 1$, where $A_0$ is its actual zero-free core and $\sigma_A$ is the actual simultaneous sorting sign. Its complete coframe is regular at $\epsilon = 0$, with leading term the core coframe. For coequal $A, B$, the core functions and sorting signs agree and their unit counts agree.*

*At a zero-free core with coframe $g$ and proper matrix $T = (U_{i-s})_{i\in I, s\in S}$, write $P^+ = [Ug]_{<M}$, with $U_0 = 0$. A restoration at a genuine vacancy $b$ has complete first coframe increment*

$$\delta^{(b)} = -P_b^+ \left( z^b - \sum_{s\in S} (T^{-1}v_b)_s z^s \right), \qquad v_b = (U_{i-b})_{i\in I}. \tag{4.2}$$

*For any rational weights $\rho_b$ in the full raw field, $\delta = \sum_b \rho_b \delta^{(b)}$ satisfies*

$$\delta_0 = 0, \qquad [U\delta]_i = 0 \ (i \in I), \qquad \delta_b = -\rho_b P_b^+ \quad \text{at each genuine vacancy } b.$$

*In particular, a nonzero mask gives a nonzero complete tangent.*

*Proof.* Collect the true unit rows and columns simultaneously. Their full block has diagonal one and off-diagonal entries of positive $\epsilon$ order. Its Schur complement has leading term the complete core matrix, with the corresponding power of $\epsilon$. This proves (4.1) and the regularity of the coframe after the core proper-corner powers are factored. It keeps the entire inverse of the unit block.

Equal whole functions have equal row counts by Lemma 2.1. Their least nonzero $\epsilon$ powers give equal unit counts and proportional core functions with their actual signs. Evaluation of actual skew Schur functions on a sufficiently large positive alphabet fixes the signs and gives whole core equality.

For one restored vacancy, the first new native equation is $\delta_b = -[Ug]_b = -P_b^+$. The old proper rows acquire the contribution $v_b\delta_b$. Solving the entire old system therefore gives the correction $P_b^+ T^{-1} v_b$, which proves (4.2). Linearity proves the displayed native equations for any weights. No other vacancy contributes at the new column $b$, and $P_b^+ \neq 0$ by the actual numerator atlas. □

**Lemma 4.2.** *Let $g$ be the complete coframe of an actual connected normalized diagram of span $M$. Let $\delta$ be any polynomial over its full raw field with $\deg\delta < M$, $\delta_0 = 0$, and $[H\delta]_i = 0$ at every original proper row. For $0 \le a \le M-1$,*

$$\mathcal{T}_{g,\delta}(z^a H) = 0 \quad \Longleftrightarrow \quad \delta = 0.$$

*If the diagram is zero-free, the same assertion with $H$ replaced by $U$, $U_0 = 0$, holds for $0 \le a \le M-2$, provided its native equations are $[U\delta]_i = 0$. No support restriction on $\delta$ is required. Vanishing modulo $z^{2M-1}$ suffices in both statements.*

*Proof.* For the first statement, set $P = [Hg]_{<M}$. The polynomial $z^a P_{<M-a}$ is coprime to $g$: $g(0) = 1$ removes $z^a$, and specializing the independent positive numerator coordinates to zero leaves $\vartheta \neq 0$.

In the zero-free case, put $P^+ = [Ug]_{<M}$. The change from the coherent numerator coordinates to $P^+$ subtracts the corresponding coframe coordinates and is invertible with $g$ fixed. The

available coefficients of $P^+$ remain independent. Its coefficient at one is available: an actual proper row at one would require column one and would be a unit. This independent coordinate survives in $z^a P^+_{<M-a}$ for $a \le M-2$. Modulo any irreducible factor of $g$, its coefficient is $z^{a+1} \neq 0$. Thus this product too is coprime to $g$; $g = 1$ causes no exception.

Write $W = H$ or $U$. A zero wedge gives

$$\delta = gR, \qquad [z^a W\delta]_{<M} = R[z^a Wg]_{<M}, \qquad R_0 = 0.$$

If $R \neq 0$, let $e = \deg R \ge 1$ and let $j$ be the largest available numerator position below $M-a$. It exists: $j \ge 0$ in the first case and $j \ge 1$ in the second. The right side has exact degree $a+j+e$. If this is at least $M$, clipping is contradicted. Otherwise $j+e$ is below $M-a$ and is greater than every available numerator position, hence is an original proper row. The left coefficient there is the required zero $[W\delta]_{j+e}$, whereas the right leading coefficient is nonzero. Thus $R = 0$. Both sides are whole polynomials of degree at most $2M-2$, giving the asserted finite precision. □

4.2. **A genuine linearized fixed-cap theorem.** The next theorem supplies the first nonzero whole-unit grade in the cell-splice comparison. Unlike Lemma 4.2, it concerns two actual polynomial observations, and its weights are constants.

Let $B$ be a zero-free connected diagram with complete matrix $M_B$. At a genuine vacancy $b$, let $a_b = (h_{x-b})_{x\in X_B}$ and $b_b = (h_{b-y})_{y\in Y_B}$, and put

$$\dot{s}^{(b)}_B = -b_b \operatorname{adj}(M_B)\, a_b. \tag{4.3}$$

If $\sigma_b$ is the actual sorted insertion sign, the complete bordered determinant is

$$s_{B+b} = \sigma_b\big(s_B + \dot{s}^{(b)}_B\big)$$

at physical unit one. More precisely, scaling positive letters by $\epsilon$ and the new unit by $\nu$ gives

$$s_{B+b} = \sigma_b\left(\nu\epsilon^r s_B + \epsilon^{r+1}\dot{s}^{(b)}_B\right), \qquad r = \operatorname{rank} M_B. \tag{4.4}$$

Thus $\dot{s}^{(b)}_B$ has the same weighted degree as $s_B$ and ordinary positive-letter degree $r+1$.

For constants $\rho_b \in \mathbb{Q}$ supported on the genuine vacancies, write $\dot{s}_B = \sum_b \rho_b \dot{s}^{(b)}_B$. On a fixed cap splice use the same weights at the same low positions. The extra row and column insertion count is even, so the normalized signs agree.

**Theorem 4.3** (Zero-free linearized fixed-cap rigidity)**.** *Let $B$ and $K$ be actual connected zero-free diagrams, and use the same constant vacancy weights on $B$ and on the actual splice $K \triangleleft B$. If*

$$\dot{s}_B = 0, \qquad \dot{s}_{K\triangleleft B} = 0$$

*as whole polynomial identities, then every $\rho_b$ is zero.*

*Proof.* Write $m$ for the span of $B$ and $v$ for that of $K$. Let $D_0$ be the complete proper matrix of $B$, $\kappa$ its signed determinant, and $G$ its complete coframe. The exact first variation $Q$ of the coframe includes both the original native columns and the new vacancy columns. The bordered solve gives

$$Q = -\sum_b \rho_b P_b \left(z^b - \sum_{s\in S_B} (D_0^{-1}v_b)_s z^s\right), \quad P_b = [\mathsf{H}G]_b, \quad v_b = (h_{i-b})_{i\in I_B}. \tag{4.5}$$

In particular $Q_0 = 0$, $\deg Q < m$, $Q_b = -\rho_b P_b$ at a vacancy, and $[\mathsf{H}Q]_i = 0$ at every original proper row.

All coefficients of $Q$ and $\kappa$ have raw ceiling below $m$. The unique top-letter coefficient of $\dot{s}_B = 0$ gives $\dot{\kappa} = 0$, and its remaining whole equation gives $[\mathsf{H}Q]_m = 0$. Differentiating the exact fixed-cap identity (3.2) then gives

$$[\mathsf{H}Q]_m = 0, \qquad [\mathsf{H}C_K Q]_{m+v} = 0. \tag{4.6}$$

If $Q$ has raw ceiling $R$, successive comparison of the genuine high-letter coefficients in the second equation gives

$$Q_j = 0 \qquad (j < m + v - \max(R, v-1)). \tag{4.7}$$

This is the same triangular removal of the constant-one coframe $C_K$ as before. If $R < v$, all coefficients vanish.

We use strong induction on $m$, uniformly over actual zero-free caps. Suppose the restoration weights in all higher strict blocks have already been removed, and let $P$ be their entire fixed proper prefix, including its crossing blocks. Write $m_0$ for its largest row, $a$ for its least column, and $S = m_0 - a$; omit an empty prefix. Let $t$ be the largest remaining proper row and $b_{\max}$ the largest remaining vacancy, using zero for an empty set. The lower coframe coefficients have raw ceiling at most $\max(t, b_{\max})$. Their full prefix compensation is

$$Q_{\text{prefix}} = -P^{-1} \left( \sum_{j<a} h_{x-j} Q_j \right)_{x \in \operatorname{rows} P}. \tag{4.8}$$

If a raw ceiling $L$ exceeds $\max(S, t, b_{\max})$, every differentiated crossing entry multiplies $Q_{x-L}$, which is zero by (4.7). The whole prefix equation therefore has zero derivative in $h_L$. Descending earns $R = \max(S, t, b_{\max})$. If $S \geq \max(t, b_{\max})$, the zero range passes $a$, so the tail and then the prefix vanish. Otherwise $R = \max(t, b_{\max}) > S$; when $R \geq v$ the zero range includes

$$Q_{m-R} = 0, \qquad Q_{x-R} = 0 \quad (x \in \operatorname{rows} P). \tag{4.9}$$

There are two next-head cases. First suppose $R = b_{\max} > t$ and write $b = R$. Every remaining baseline proper row is below $b$; its entry in the restored column is genuinely zero. Also $P_b = h_b +$ a function of lower raw letters. Differentiating (4.5) and (4.8) in $h_b$, all crossing derivatives multiply the zeros in (4.9). The surviving pulse is $-\rho_b$ at the new column and its full prefix correction. Differentiating the whole top equation gives

$$\rho_b \left( h_{m-b} - H_{m,\operatorname{cols} P} P^{-1} H_{\operatorname{rows} P, b} \right) = 0. \tag{4.10}$$

The numerator in parentheses has actual rows $\{m\} \cup \operatorname{rows} P$ and columns $\operatorname{cols} P \cup \{b\}$. On $[b, m)$ its height is one plus the prefix height, with the two genuine singleton ends. It is therefore an actual connected skew diagram with nonzero Schur function. Its full denominator is $\det P$. Consequently $\rho_b = 0$. This also handles the case with no proper block left.

Now suppose $R = t > b_{\max}$. Let $E$ be the next actual zero-free strict block, with head $t$ and least column $a_0 > 0$. The complete proper matrix has triangular form

$$\begin{pmatrix} P & B_{PE} & B_{PL} \\ 0 & E & B_{EL} \\ 0 & 0 & L \end{pmatrix}.$$

The inverse column selected by row $t$ has no component in the lower suffix $L$. After all suffix factors cancel as a whole, its top row-replacement quotient is, with its actual sign, $s_N/(\det P\, s_E)$, where

$$X_N = \{m\} \cup \operatorname{rows} P \cup (\operatorname{rows} E \setminus \{t\}), \qquad Y_N = \operatorname{cols} P \cup \operatorname{cols} E.$$

Differentiate the whole equation $[\mathsf{H}Q]_m = 0$ in $h_t$. The direct term and the differentiated crossings vanish by (4.9). A restoration in $L$ cancels in the complete triangular quotient. A restoration in $E$ affects both its determinant and the whole numerator $N$, with the same weight. Multiplication by all actual denominators gives precisely

$$s_E \dot{s}_N - s_N \dot{s}_E = 0. \tag{4.11}$$

The numerator is connected: the original prefix interval joins the true strict block with its singleton end caps. The lists

$$X_{K_E} = \{m - t\} \cup (\operatorname{rows} P - t), \qquad Y_{K_E} = (\operatorname{cols} P - t) \cup \{0\}$$

define an actual connected zero-free cap. After translating by $a_0$, one has literally $N = K_E \triangleleft E$. Its span $m - a_0$ exceeds the child span $t - a_0$.

Thus $s_E, s_N$ are coprime connected functions, and (4.11) implies $s_E \mid \dot{s}_E$. Since $E$ is zero-free, (4.4) makes $\dot{s}_E$ have the same positive weighted degree as $s_E$ and ordinary positive-letter degree one greater. A polynomial quotient would have weight zero and ordinary degree one, which is impossible in positive-weight letters. Hence $\dot{s}_E = 0$, and then $\dot{s}_N = 0$. The induction hypothesis applies to the smaller child $E$ and the actual zero-free cap $K_E$. It removes all restoration weights in $E$. The sorted insertion parity changes by an even number, so these are the original weights.

Extend the fixed prefix and repeat. Each remaining vacancy is in a genuine gap or a genuine strict block, so the process terminates. Equivalently, at its conclusion $Q = 0$, and $Q_b = -\rho_b P_b$ at each previously absent native column forces every $\rho_b = 0$, because the actual feasible numerator $P_b$ is nonzero. □

### 4.3. **The first cell-splice grade and finite shifts.**

**Lemma 4.4.** *Let $V$ be an actual connected diagram that is not its normalized rotation. At physical normalization put*

$$J_2 = s_{V \triangleleft \mathrm{CELL}} - s_{\mathrm{CELL} \triangleleft V}.$$

*Let its zero-free core have $k$ rows and let $u$ be the number of actual units. Scale the positive alphabet by $\varepsilon$, retaining the coherent unit $\nu \neq 0$. Then $J_2$ is nonzero. Its first nonzero grade is*

$$\lambda = k \quad \text{if the core is not self-rotating,} \qquad \lambda = k + 1 \quad \text{if the core is self-rotating.}$$

*The full absolute adjugate difference between $V$ and its rotation has no grade below $\mu = \lambda - 1$. Consequently every supplied current formed by a single positive raw contraction of that whole array has order at least $\lambda$.*

*Proof.* The full simultaneous unit Schur complement gives, with the actual sorting sign $\tau$,

$$s_V = \tau \nu^u \varepsilon^k \left( s_B + \frac{\varepsilon}{\nu} \sum_{b \in U_V} \dot{s}_B^{(b)} + O(\varepsilon^2) \right). \tag{4.12}$$

It applies to both literal cell splices with all unit positions retained. If $B$ is not self-rotating, rotating one of the splice observations turns vanishing of its leading $J_2$ into equality of the same literal-cell extensions of $B$ and $B^*$. They already have equal whole functions. Theorem 3.2 would give $B = B^*$, a contradiction. Thus grade $k$ is nonzero.

If $B = B^*$, the constant core observations coincide. Since $V \neq V^*$, the actual weights $\rho_b = \mathbf{1}_{U_V}(b) - \mathbf{1}_{U_V}(M - b)$ are not all zero. The whole equality $s_V = s_{V^*}$ at the next grade gives $\dot{s}_B = 0$. If the grade-$k+1$ cell-splice difference vanished, the same weights and sorting signs would give $\dot{s}_{\mathrm{CELL} \triangleleft B} = 0$. Both the child and this cap are zero-free, so Theorem 4.3 would make every $\rho_b = 0$. Thus this next grade is nonzero.

For completeness, place the common unit flags last in both orders in the whole adjugate. Its first possible grade is $k - 1$; at that grade only core-to-core coefficients remain, and the entire absolute array is $\tau \nu^u \operatorname{adj}(\mathrm{JT}(B))$. For a self core the corresponding entire arrays agree, including the sorting sign obtained from the equal whole determinants. The next grade is attained at any actual unit position present on one side and absent on its rotation: delete that row and column zero and match all remaining actual units. The remaining weak flags have nonnegative heights and disjoint lists, hence a nonzero positive-letter determinant of degree $k$. The other array has no entry at that actual row. If the core is not self, the analogous unmatched core boundary cofactor is already nonzero at grade $k-1$. This proves the asserted whole-array orders. Multiplication by one positive raw letter raises the order by one; the complete contraction cannot create a lower grade. This last conclusion refers to the supplied contraction of the whole array, not to independent observations of its tied entries. □

## 5. Finite stationary separation

The exponent in this section is an integer at least two, not a composition period. The coframes belong to actual connected diagrams with equal whole Schur functions, at their supplied common span.

5.1. **Highest-component matching.** The highest strict proper component has a head $m$, least column $a$, and span $L = m - a$. The inverse column selected by its head has the form

$$-z^a g/k, \qquad k = [Hg]_L, \tag{5.1}$$

where $g$ is the component's own right coframe. This is Cramer's rule applied to the complete block-triangular proper matrix: the selected inverse column has zero lower-suffix entries, and all lower suffix factors cancel as a whole. For a true unit, $L = 0$, $g = 1$, $k = \vartheta$.

**Lemma 5.1.** *For an actual connected component of span $L \geq 2$ and any integer $p \geq 2$,*

$$[H^p g]_L = p\vartheta^{p-1}k + \alpha, \qquad k = H_L + k_0, \tag{5.2}$$

*where $k_0$ and $\alpha$ are independent of $H_L$, and $\alpha \neq 0$.*

*Proof.* The independence of $H_L$ is immediate from the complete proper solve. On the child's neutral top $k = 0$, the remainder becomes $[P^p/g^{p-1}]_L$, where $P = [Hg]_{<L}$ and $P_0 = \vartheta$. Use the actual independent coordinates of Lemma 2.2. If $g \neq 1$, let $b$ be its least native column and write $L = nb + j$, $0 \leq j < b$. Actual feasibility makes $P_1, \dots, P_{b-1}$ available. For $j = 0$, the coefficient of the independent monomial $g_b^n$ is $(-1)^n\binom{p+n-2}{n}\vartheta^p$; for $j > 0$, that of $P_j g_b^n$ is $(-1)^n p\binom{p+n-2}{n}\vartheta^{p-1}$. Both are nonzero. If $g = 1$, the component is a row. At $L = 2$, the coefficient of $P_1^2$ is $\binom{p}{2}\vartheta^{p-2}$, and at $L \geq 3$, that of $P_1 P_{L-1}$ is $p(p-1)\vartheta^{p-2}$. This proves nonvanishing, including the two-cell row at $p = 2$. □

**Lemma 5.2.** *Let $A, B$ be actual connected coequal diagrams of span $M$, both having proper row $m = M - 1$. Put $\Delta = C - G$. Suppose, for an integer $p \geq 2$,*

$$[H^p\Delta]_m = 0, \qquad \partial_{H_m}\big([H^p\Delta]_M - \Delta_1[H^p G]_m\big) = 0.$$

*Then their highest strict proper components agree literally in position. Moreover $\partial_{H_m}\Delta = 0$ and $\Delta_1 = 0$.*

*Proof.* Put $y = H_m$. Let the two top components have least columns $a, b$, spans $L = m - a$, $L' = m - b$, coframes $g, c$ and quotients $k_E, k_F$. By (5.1),

$$\partial_y G = -z^a g/k_E, \qquad \partial_y C = -z^b c/k_F.$$

The coefficient of $y$ in the first observation gives

$$\frac{[H^p g]_L}{k_E} = \frac{[H^p c]_{L'}}{k_F}. \tag{5.3}$$

The explicit derivative of $H^p$ multiplies $\Delta_0 = 0$. For $L \geq 2$, Lemma 5.1 gives a genuine pole in the independent top letter $H_L$. If $L > L'$, the right side is independent of that letter, an impossibility. The remaining unequal case, $L = 1, L' = 0$, compares $p\vartheta^{p-1}$ with $\vartheta^{p-1}$ and is also impossible. Interchanging the diagrams gives $L = L'$ and $a = b$.

For $L = 0$, both components are the same true unit: $g = c = 1$, $k_E = k_F = \vartheta$. For $L = 1$, both are the same cell: $g = c = 1$, $k_E = k_F = H_1$. In either case $a = b$, and the complete selected inverse columns, including their zero lower suffixes, agree. Hence $\partial_y\Delta = 0$. Differentiating the whole coequal identity gives

$$0 = \partial_y[H\Delta]_M = [z^{M-1}\Delta]_M + [H\partial_y\Delta]_M = \Delta_1.$$

This closes both small cases with their actual unit or cell denominator.

For $L \geq 2$, write $k_E = H_L + k_{E,0}$ and $k_F = H_L + k_{F,0}$. Cross-multiplication in (5.3) first gives equal nonzero remainders and then $k_{E,0} = k_{F,0}$. Thus $k_E = k_F =: k$. Connected primeness and coprimality with the full proper corners identify the actual child functions and their signed corners; primitive normalization fixes the scalar.

Assume $L \geq 2$, and put $\delta = c - g$, $A_p = [H^p g]_L$, $B_p = [H^p\delta]_{L+1}$, $V = [H\delta]_{L+1}$. These expressions are independent of $y$. Also $\Delta_1$ is $y$-independent, since $\delta_0 = 0$ and $a \geq 1$. The second supplied observation gives $B_p = \Delta_1 A_p$. Differentiating the whole equality $[H\Delta]_M = 0$ gives $V = \Delta_1 k$. In particular,

$$kB_p = VA_p. \tag{5.4}$$

Set $x = H_L$ and write

$$k = x + k_0, \quad V = x\delta_1 + v_0, \quad A_p = p\vartheta^{p-1}k + \alpha, \quad B_p = p\vartheta^{p-1}V + \beta.$$

All subscript-zero terms and $\alpha, \beta$ are independent of $x$, and $\alpha \neq 0$. Equation (5.4) becomes $\beta k = \alpha V$. Comparing $x$ and then its constant coefficient gives $V = \delta_1 k$.

Let $\kappa_*$ be the common signed child corner. The difference of the two actual same-cell splice functions is the polynomial

$$B_{\mathrm{cell}} = \kappa_*([Hg]_{L+1} - [Hc]_{L+1}) = -s_E\delta_1.$$

The product $\kappa_*\delta_1$ is polynomial. Since $\gcd(s_E, \kappa_*) = 1$, the connected function $s_E$ divides $B_{\mathrm{cell}}$. Both splices add one cell without increasing the child row count, so $B_{\mathrm{cell}}$ has positive row degree at most that of $s_E$. A nonzero quotient would have positive weighted degree one and therefore positive ordinary degree, contradicting Lemma 2.1. Hence $B_{\mathrm{cell}} = 0$. The actual children have equal whole functions and equal same-cell splice functions. Theorem 3.2 identifies them literally. Thus $g = c$, $\partial_y\Delta = 0$, and $\Delta_1 = 0$ follows from $V = \Delta_1 k$. □

**Lemma 5.3.** *Suppose two actual coequal diagrams have their highest strict proper components literally matched in position, with least column $a$. If their complete coframes differ, then* $1 \leq \mathrm{ord}_z(C - G) \leq a - 2$.

*Proof.* Let $P$ be the complete common top matrix and $\delta = [C - G]_{<a}$. Subtraction of its full native equations gives

$$((C - G)_s)_{s\in\mathrm{cols}\,P} = -P^{-1}\left(\sum_{j<a} H_{x-j}\delta_j\right)_{x\in\mathrm{rows}\,P}.$$

Thus a nonzero difference starts below $a$. If it started at $a - 1$, the exact native support would identify all columns below $a-1$, and the numerator-zero test would identify all rows there. The common top identifies both flags at or above $a$. Equal total row counts then determine row membership at $a - 1$, and equality of the proper row and column counts determines column membership there. All flags would agree, contradicting the distinct coframes. □

5.2. **A finite pole calculation.** The potential in the following theorem is an arbitrary fixed rational function. Its value is not calibrated in advance. The proof uses only coefficients already present at the stated finite horizon.

**Theorem 5.4** (One-power stationary separation)**.** *In the domain of Lemma 2.4, let $p \geq 2$ be an integer and $r(x) \in K(x)$. If*

$$R(z) := \mathcal{T}_{G,C}(zH^p) + zDr(H) = 0 \pmod{z^{2M}}, \tag{5.5}$$

*then $A = B$ literally. Conversely, literal equality makes the identity hold for every such rational potential.*

*Proof.* Suppose $A \neq B$, and put $m = M - 1$ and $\Delta = C - G$. Because $zDr(H)$ has order at least $M + 2$, the first two retained coefficients give

$$E_0 = [H^p\Delta]_m = 0, \qquad E_1 = [H^p\Delta]_M - \Delta_1[H^pG]_m = 0. \tag{5.6}$$

Indeed, direct subtraction of the two discarded tails gives

$$[\mathcal{T}(zH^p)]_M = -E_0, \qquad [\mathcal{T}(zH^p)]_{M+1} = -E_1 - G_1E_0.$$

The corresponding coefficient is retained also when $M = 3$.

We first earn the maximal-row incidence. If exactly one proper row set contains $m$, the full derivative of $E_0$ in $y = H_m$ is, up to sign,

$$\frac{[H^pg]_L}{k},$$

where $g$ and $k = [Hg]_L$ belong to its actual highest strict proper component. This is nonzero: for a positive component its numerator has coefficient $p\vartheta^{p-1}$ at $H_L$, and a true unit gives $\vartheta^{p-1}$.

The inverse-column calculation is (5.1). If neither proper row set contains $m$, the complete coframes are independent of $y$, and differentiation of (5.5) yields

$$R_y = -z^M \Delta r(H) \pmod{z^{2M}}.$$

The derivative of the potential itself starts beyond the retained window. If $r \neq 0$, then $r(\vartheta) \neq 0$ because $\vartheta$ is transcendental, and the first term of this product is visible. Thus $r = 0$. Now $\mathcal{T}(zH^p) = 0$ is a whole polynomial identity. Coprimality of $G, C$ implies $[zH^pG]_{<M} = AG$ with $\deg A \leq M-2$. Division by $G$, whose constant is one, gives $A = zH^p \pmod{z^M}$. Its degree-$M-1$ coefficient contains the nonzero independent-letter term $p\vartheta^{p-1}H_{M-2}$, a contradiction. Consequently both proper row sets contain $m$.

By Lemma 5.2, (5.6) identifies their highest strict proper components literally in position. Write $a$ for the common least column, $L = m - a$ for its span, $g$ for its coframe, and $k = [Hg]_L$ for its quotient. A true unit has $L = 0$, $g = 1$, $k = \vartheta$. The complete inverse columns agree:

$$G = G_0 + yv, \quad C = C_0 + yv, \quad v = -\frac{z^a g}{k}, \quad \partial_y \Delta = 0. \tag{5.7}$$

The support bound in Lemma 5.3 gives

$$d := \operatorname{ord}_z \Delta \leq a - 2. \tag{5.8}$$

Put $h = H - yz^m$. Differentiating the whole operator, including its discarded tails, gives

$$D_y = -\mathcal{T}_{\Delta,v}(h) - z^m \Delta.$$

The second term is the parent correction. Since the whole $D_y$ has order at least $M+1$, both the explicit $r'(H)$ term and the replacement of $r(H)$ by $r(h)$ lose only degrees at least $2M+1$. Substituting (5.7), and using $M - a = L + 1$, therefore gives

$$\begin{aligned} R_y = \frac{z^{a+1}}{k}\big\{&\Delta[h^p g]_{<L} - g[h^p \Delta]_{<M-1} \\ &+ r(h)\big(\Delta[hg]_{<L} - g[h\Delta]_{<M}\big)\big\} \pmod{z^{2M}}. \end{aligned} \tag{5.9}$$

In this expression the term $kz^L\Delta$ from the first clipping has canceled the parent correction. It has not been omitted.

At inner degree $L + d$, the two products containing $\Delta$ are unclipped, since $L + d \leq M - 3$. Thus

$$[z^{M+d}]R_y = -\Delta_d \left( \frac{[h^p g]_L}{k} + r(\vartheta) \right). \tag{5.10}$$

This coefficient lies strictly below $2M$.

*A component of span $L \geq 2$.* Lemma 5.1 writes

$$[h^p g]_L = p\vartheta^{p-1}k + \alpha, \qquad k = H_L + k_0, \qquad \alpha \neq 0,$$

where $\alpha, k_0$ are independent of $H_L$. Divide (5.10) by the nonzero rational function $\Delta_d$; there is no need for $\Delta_d$ itself to be independent of $H_L$. The $H_L$ coefficient gives $r(\vartheta) = -p\vartheta^{p-1}$ and the constant coefficient then gives $\alpha = 0$. This is impossible.

*The cell, $L = 1$.* Here $g = 1$, $k = H_1$, $a = M - 2$. Equation (5.10) gives $r(\vartheta) = -p\vartheta^{p-1}$, hence $r(x) = -px^{p-1}$ as a rational identity. With

$$Q_1 = [h^p\Delta]_{<M-1} - \vartheta^p\Delta - ph^{p-1}\big([h\Delta]_{<M} - \vartheta\Delta\big),$$

equation (5.9) becomes $R_y = -z^{M-1}Q_1/H_1$. Below degree $M-1$, $Q_1$ is the product of $\Delta$ with

$$(1-p)h^p + p\vartheta h^{p-1} - \vartheta^p.$$

The constant and linear terms vanish. The quadratic coefficient is

$$-\binom{p}{2}\vartheta^{p-2}H_1^2.$$

Since $d \le M-4$, its first surviving product coefficient is retained, and

$$[z^{M+d+1}]R_y = \binom{p}{2}\vartheta^{p-2}H_1\Delta_d \neq 0. \tag{5.11}$$

This excludes the cell case, including $p = 2$.

*The true unit,* $L = 0$. Equation (5.10) now gives $r(x) = -x^{p-1}$. Both original diagrams have a true unit at $m$. Normalize coherently and set $H_\epsilon = 1 + \epsilon U$, where $U_0 = 0$ and its positive coefficients remain independent. The complete unit Schur complement of Lemma 4.1 gives

$$G_\epsilon = g + \epsilon G_1 + \cdots, \qquad C_\epsilon = c + \epsilon C_1 + \cdots,$$

where $g, c$ belong to their actual zero-free cores. Those cores have equal whole functions and both omit row and column $m$. The true restored $m$-columns have first coefficients $-[Ug]_m$ and $-[Uc]_m$. Hence the coefficient of $\epsilon$ in $E_0 = 0$ is

$$(p-1)([Uc]_m - [Ug]_m) = 0. \tag{5.12}$$

Shortening only the cores' top beta from $M$ to $m$ leaves their proper matrices unchanged and produces actual connected zero-free diagrams. Their whole functions agree by (5.12); adjoining the same literal cell on top recovers the two original cores, whose whole functions also agree. The nonlinear fixed-cap result in Theorem 3.2 identifies the cores literally, so $g = c$.

If the original unit masks agree, the original diagrams are already equal. Otherwise their full first difference $\delta = C_1 - G_1$ is nonzero: at a differing genuine vacancy $b$ it has coefficient $-\rho_b[Ug]_b$, where $\rho_b$ is the difference of the two restoration indicators. All compensating old-column terms are included in Lemma 4.1. In particular, $\delta_0 = 0$ and $[U\delta]_i = 0$ at every core proper row.

Write the alternating pair of coframes as $G_\epsilon \wedge C_\epsilon = \epsilon W_1 + \epsilon^2 W_2 + \cdots$, where $W_1 = g \wedge \delta$, and set $D_1 = \mathcal{T}_{W_1}(U)$. Here $\mathcal{T}$ depends linearly on the alternating pair; this notation only groups complete coefficients. Both original maximal row equations give the entire identity

$$\mathcal{T}_{G_\epsilon, C_\epsilon}(zH_\epsilon) = z\mathcal{T}_{G_\epsilon, C_\epsilon}(H_\epsilon). \tag{5.13}$$

The $\epsilon^2$ coefficients of this identity and of (5.5), respectively, are

$$\mathcal{T}_{W_2}(z) + \mathcal{T}_{W_1}(zU) - zD_1 = 0, \qquad \mathcal{T}_{W_2}(z) + p\mathcal{T}_{W_1}(zU) - zD_1 = 0.$$

Their difference is $(p-1)\mathcal{T}_{g,\delta}(zU) = 0$. These are whole polynomials of degree below $2M$. Finite zero-free coframe injectivity, Lemma 4.2, forces $\delta = 0$, a contradiction. Every incidence and highest component case has now been exhausted. □

### 5.3. **The whole moving potential.**

**Corollary 5.5.** *Retain the domain of Lemma 2.4: the actual connected diagrams $A, B$ with $s_A = s_B$, their complete coframes $G, C$, the defect $D = \mathcal{T}_{G,C}(H)$, and the same supplied $M$. Let $\mathbb{F} \supseteq K(\vartheta)$ be the neutral coefficient field, with all retained positive coarse raw letters algebraically independent over $\mathbb{F}$. Write* $\mathbf{H}$ *for that finite tuple of letters. Suppose the supplied whole moving homomorphism is*

$$\mathcal{R} : \mathbb{F} \longrightarrow \mathbb{F}(\mathbf{H})[z]/(z^{2M}), \qquad \mathcal{R}|_K = \mathrm{id}, \quad \mathcal{R}(\vartheta) = H, \quad \mathcal{R}(a) \equiv a \pmod{z} \quad (a \in \mathbb{F}).$$

*It acts on every dependent rational coordinate, including complete inverse entries; rational substitution is legitimate because every nonzero denominator has its original nonzero constant term. If $c \in K^\times$, $p \ge 2$ is an integer, and $\psi_0 \in \mathbb{F}$ satisfies*

$$c\mathcal{T}_{G,C}(zH^p) + zD\mathcal{R}(\psi_0) = 0 \pmod{z^{2M}}, \tag{5.14}$$

*then $A = B$ literally.*

*Proof.* For a distinct pair put $\ell = \mathrm{ord}_z D \le 2M-2$. Its retained coefficient at degree $\ell+1$ gives

$$\frac{\psi_0}{c} = -\frac{[z^{\ell+1}]\mathcal{T}_{G,C}(zH^p)}{D_\ell}.$$

The left side belongs to $\mathbb{F}$, the right side to $K(\vartheta, H_1, \dots, H_{M-1})$. Raw independence gives

$$\mathbb{F} \cap K(\vartheta, H_1, \dots, H_{M-1}) = K(\vartheta).$$

Thus $\psi_0/c = r(\vartheta)$ for a fixed rational function $r \in K(x)$. Applying the whole homomorphism, which fixes $c$, gives $\mathcal{R}(\psi_0)/c = r(H)$. This contradicts Theorem 5.4. The argument does not assume that $D_{M+1}$ is nonzero. □

The coefficient at $2M-1$ cannot be discarded: it can be the first one at which a potential occurs when $\operatorname{ord} D = 2M-2$. Likewise an amplitude can be divided out only when it belongs to the fixed, raw-independent field. A coefficient of the fine field must move under $\mathcal{R}$, just as the returned coordinate does. An application must establish that its amplitude is in the fixed field before invoking this corollary.

**Example 5.6.** For $A = (2,1)$ and $B = (2,2)/(1)$, the full coherent native matrices are

$$\mathrm{JT}(A) = \begin{pmatrix} H_2 & H_3 \\ \vartheta & H_1 \end{pmatrix}, \qquad \mathrm{JT}(B) = \begin{pmatrix} H_1 & H_3 \\ \vartheta & H_2 \end{pmatrix}.$$

Both determinants are $H_1H_2 - \vartheta H_3$, and $M = 3$. Their proper native equations are $\vartheta G_1 = -H_1$ and $\vartheta C_2 = -H_2$, respectively. Consequently

$$G = 1 - \frac{H_1}{\vartheta} z, \qquad C = 1 - \frac{H_2}{\vartheta} z^2, \qquad D = \frac{H_2(\vartheta H_2 - H_1^2)}{\vartheta^2} z^4.$$

Thus $\operatorname{ord} D = 4 = 2M-2$, and the first coefficient containing the moving potential in $zD\mathcal{R}(\psi_0)$ has degree $5 = 2M-1$. This explains the terminal coefficient in this potential-descent argument; it does not assert that every weaker recognition test fails.

The specialization $u = H$ of (2.3) gives $\mathcal{T}(zH) - zD$. It vanishes exactly when the whole cap polynomial in that formula vanishes. The two actual maximal-row equations in the unit argument supply the separate scalar zeros, which are sufficient but not necessary for arbitrary coframes.

## 6. Unshifted opposite-end exclusion

For a board of span $M$, use the coherent clock $\vartheta \neq 0$, independent positive letters $H_i$, and its two complete coframes $G, C$, with constant one. Put

$$P = [HG]_{<M}, \quad Q = [HC]_{<M}, \quad D = GQ - CP, \quad \mathcal{T}_{A,B}(F) = A[FB]_{<M} - B[FA]_{<M}.$$

Every truncation in this section is literal. In particular

$$\mathcal{T}_{G,C}(zH) - zD = -z^M\{G[HC]_{M-1} - C[HG]_{M-1}\};$$

its vanishing is equivalent to the vanishing of the displayed *whole polynomial*. Separate scalar-cap zeros are sufficient, not necessary for arbitrary coframes.

The equation below is unshifted. It is not (5.5): moving $z$ through the finite operator creates the whole cap term in (2.3).

**Lemma 6.1.** *Let $V$ be an actual connected diagram of span $M \geq 3$, and let $G, C$ be its complete right and left coframes, equivalently the right coframes of $V, V^*$, in the same coherent raw alphabet. Retain every proper component and every actual unit. For an integer $p \geq 2$, the identity*

$$\mathcal{T}_{G,C}(H^p) = p[H^{p-1}D]_{<2M}$$

*at precision $2M$, including degree $2M-1$, implies $V = V^*$. This assertion concerns opposite ends of one actual board, not arbitrary coequal coframe pairs.*

*Proof.* Suppose $V \neq V^*$. First, a connected $k$-row determinant has ordinary positive-letter degree exactly $k$. Its degree-$k$ component is obtained by replacing all units by zero. The substitution $H_i \mapsto h_{i-1}$, $i > 0$, turns it into the weak skew determinant obtained by removing one cell from every row, which is nonzero. Thus a nonzero polynomial of degree at most $k$ and

weight greater than the board's area cannot be divisible by its connected prime. The actual cell-cap difference $J_2 \neq 0$ from Lemma 4.4 has these degree and surplus-weight properties. Hence it survives the genuine top-neutralization. The complete coframe formula yields

$$[z^{M+1}]D = -J_2|_{e=0}/\kappa \neq 0 \tag{6.1}$$

for an actual nonself board. This specific conclusion uses the cell-cap theorem, not merely $D \neq 0$. We will apply it to actual connected components in their own fields.

Set $m = M - 1$, $y = H_m$, and $H = h + yz^m$, with $h_m = 0$. Let $I$ and $S$ be the proper row and column sets, and let $\Pi$ be the complete proper inverse, independent of $y$. Define

$$a = -\sum_{c\in S} \Pi_{c,m} z^c, \qquad b = -\sum_{i\in I} \Pi_{1,i} z^{M-i}.$$

A missing native port means the zero vector. Then $G = G_0 + ya$, $C = C_0 + yb$, where the subscript zero means $y = 0$. Write $\Delta = G_0 - C_0$, $W_0 = G_0 \wedge C_0$, $W_1 = a \wedge C_0 + G_0 \wedge b$, $W_2 = a \wedge b$. For a wedge $W = A \wedge B$, put

$$\mathcal{E}_W^M = \mathcal{T}_{A,B}^M(h^p) - ph^{p-1}\mathcal{T}_{A,B}^M(h),$$

always at precision $2M$. The complete coefficients of the defect are

$$\begin{aligned}
[y^0]\mathcal{E} &= \mathcal{E}_{W_0}^M, \\
[y^1]\mathcal{E} &= \mathcal{E}_{W_1}^M + p(\vartheta^{p-1} - h^{p-1})z^m\Delta, \\
[y^2]\mathcal{E} &= \mathcal{E}_{W_2}^M + p(\vartheta^{p-1} - h^{p-1})z^m(a-b).
\end{aligned}$$

To verify them, note $\mathcal{T}_{W_0}(z^m) = z^m\Delta$, $\mathcal{T}_{W_1}(z^m) = z^m(a-b)$, $\mathcal{T}_{W_2}(z^m) = 0$. Every coefficient in $y$ of the entire $D$ has order at least $M+1$, so an additional explicit $yz^m$ in the derivative factor is beyond $2M-1$. This justifies the expansion without assigning zero to any tied summand.

If neither extreme port occurs, $a = b = 0$, and the linear equation forces $\Delta = 0$: its nonzero scalar factor has order one, and $\deg\Delta < M$. Full support would make the board self. If exactly one extreme occurs, the quadratic equation gives the same contradiction with $a - b \neq 0$. Thus both $m \in I$ and $1 \in S$ must occur.

Suppose first the entire proper corner is connected. Translate it by $-1$, obtaining an actual nonself child $E$ of span $L = M - 2$. Let $g, c$ be its two own coframes and $k = e_E/\kappa_E = H_L + k_0$. Full Cramer normalization gives $a = -zg/k$, $b = -zc/k$. Divide the quadratic equation by $z^2/k^2$; its clipping window is $L+1$, and its precision is $2L+2$. Since $[hg]_L = [hc]_L = k$, retaining both cap terms gives exactly

$$\mathcal{E}_{g,c}^L + z^L(g\alpha_c - c\alpha_g) = 0 \pmod{z^{2L+2}}, \quad \alpha_c = [h^p c]_L - p\vartheta^{p-1}k, \quad \alpha_g = [h^p g]_L - p\vartheta^{p-1}k.$$

Both caps are independent of $H_L$. The coefficient of $H_L z^{2L+1}$ is $-p(p-1)\vartheta^{p-2}[z^{L+1}]D_E$, nonzero by (6.1). This is a contradiction.

It remains to treat a disconnected corner, including unit components. Let $A$ be the least column of its top strict component, whose head is $m$, and let $M - B$ be the greatest row of its bottom component, whose least column is one. Strict separation gives $A + B > M$. Let $E$ be the top component translated by $-A$, of span $L = m - A$, and $F$ the rotated bottom component, of span $K = m - B$. For positive components use their whole coframes $g, c$ and $k_E = e_E/\kappa_E$, $k_F = e_F/\kappa_F$. For a unit use $g = 1, k_E = \vartheta$, or $c = 1, k_F = \vartheta$, and span zero. The two full endpoint inverse columns are exactly

$$a = -z^A g/k_E, \qquad b = -z^B c/k_F.$$

This follows from the actual triangular inverse; it does not delete any intervening component or identify coequal poles.

For any such component define

$$\alpha_{E,p} = [(H^p - p\vartheta^{p-1}H)g]_{L+1} - p(p-1)\vartheta^{p-2}H_1 k_E.$$

This rational function is nonzero except when $p = 2$ and $E$ is the literal two-cell row. For a unit it is $-p(p-1)\vartheta^{p-1}H_1$; for a cell it is $-\binom{p}{2}\vartheta^{p-2}H_1^2$. For $L \geq 2$ it is independent of $H_L$. If it were zero, solving the genuine child top equation $k_E = 0$ would give $[z^{L+1}]P^p/g^{p-1} = 0$ in the full

own atlas. Let $d$ be the least native column and write $L+1 = ad+b$, $0 \le b < d$. The coordinate monomial $g_d^a \vartheta^p$ when $b = 0$, or $P_b g_d^a \vartheta^{p-1}$ when $b > 0$, has a nonzero binomial coefficient. The required $P_b$ is an actual available coordinate because $b < d$. Independence forbids cancellation. When there is no native column, the component is a row. For $L \ge 3$ use $P_2 P_{L-1} \vartheta^{p-2}$, with the square interpreted when $L = 3$; for $L = 2, p \ge 3$ use $P_1^3 \vartheta^{p-3}$. The remaining $L = 2, p = 2$ is exactly the stated exception.

Suppose $A < B$, the other inequality following by reflection. At degree $M + A$ the full quadratic equation gives $\alpha_{F,p} = 0$. Indeed its coefficient is

$$-a_A\{[H^p b]_M - p\vartheta^{p-1}[Hb]_M + p(p-1)\vartheta^{p-2}H_1\},$$

which becomes that equation after the actual denominator is inserted. Thus $p = 2$, $F = \mathrm{Row}_2$, $B = M-3$, and $b = -z^{M-3}/H_2$. Put $R = H - \vartheta - H_1 z$. Dividing the entire quadratic equation gives

$$R^2 a + (H^2 a)_{\ge M} - 2H(Ha)_{\ge M} + 2H_2(\vartheta - H)z^2 b = 0 \pmod{z^{M+3}}.$$

Below $M$ it forces $A \ge M-4$; hence $A = M-4$ and $L = 3$. At degree $M$ its coefficient is $(H_1^2 g_2 - 2H_2^2)/k_E$. Actual span-three flags give, exhaustively,

$$g_2 = 0, \qquad -H_2/\vartheta, \qquad (H_1^2 - \vartheta H_2)/\vartheta^2.$$

None makes the numerator zero. Unequal endpoint spans are impossible.

If $A = B$, factor $z^{2A}/(k_E k_F)$ from the full quadratic equation. Its precision is $2L+2$. Keeping both caps as above gives

$$\mathcal{E}^L_{g,c} + z^L(g\alpha_c - c\alpha_g) = 0 \pmod{z^{2L+2}},$$

now with the separate $k_E, k_F$ in their definitions. The coefficient of $H_L z^{2L}$ forces $k_E = k_F$; the next coefficient forces $[z^{L+1}]\{g[Hc]_{<L} - c[Hg]_{<L}\} = 0$. Connected primality, primitivity and the row-degree calculation give $e_E = e_F$ and $\kappa_E = \kappa_F = \kappa$. The second equation says that the actual same-cell cap difference is $e_E(g_1 - c_1)$. Since $\kappa(g_1 - c_1)$ is polynomial and $(e_E, \kappa) = 1$, the connected prime divides this cap difference. The positive-degree bound proved above forces it to vanish. The actual fixed-cap theorem then gives $E = F$ literally. At span zero both components are already units; at span one both are cells. Thus in every case $a = b$.

Finally $\Delta = G - C \ne 0$ is independent of $y$. Its order is at most $\lfloor M/2 \rfloor$: agreement of $G, C$ and therefore $P, Q$ through that index would identify all low native supports and, by reflection, all high ones, making the board self. Since $2A > M$, this order is below $A$. The first nonzero coefficient of the complete linear equation now forces $\alpha_{E,p} = 0$. Hence $p = 2$, both endpoint components are $\mathrm{Row}_2$, and $A = M - 3$. The same full linear equation below degree $M$ becomes $(H - \vartheta - H_1 z)^2 \Delta = 0 \pmod{z^M}$; thus $\operatorname{ord} \Delta \ge M - 4$. For $M \ge 9$ this contradicts the half-window bound. For $M = 7, 8$, the endpoint components have rows $M-1, 3$ and columns $M-3, 1$. Every intervening native position lies in $[4, M-4]$, which is empty or consists only of the possible unit four. Each balanced complete flag list is literally self. This is again a contradiction.

These cases exhaust the complete strict-component decomposition and exclude every exponent $p \ge 2$ on an actual nonself board. □

## 7. Concluding remarks

One determinant can lose the positions of its boundary data. A complete coframe retains them through native support, numerator zeros, and inverse poles. The same literal cap makes two whole functions sufficient for reconstruction, while the separate constant-weight theorem controls the first variation left after a zero-free core has been identified.

Finite stationary separation uses the complete identity at the original span, including its terminal coefficient. The whole-field corollary requires raw independence and a homomorphism moving every dependent coordinate; it does not permit a moving amplitude to be divided out as a constant. The unshifted theorem requires the two ends of one actual board. None of these results classifies arbitrary rational scalar observations or supplies a marked composition chart from factor functions. Such geometric realization is a separate question.

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## References

[1] F. Barekat, V. Reiner, and S. van Willigenburg, *Corrigendum to "Coincidences among skew Schur functions" [Adv. Math. 216 (1) (2007) 118–152]*, Adv. Math. **220** (2009), no. 5, 1655–1656, DOI: 10.1016/j.aim.2008.10.010.

[2] V. Reiner, K. M. Shaw, and S. van Willigenburg, *Coincidences among skew Schur functions*, Adv. Math. **216** (2007), no. 1, 118–152, DOI: 10.1016/j.aim.2007.05.006.

[3] J. Shields, *Boundary reconstruction and cofactor rigidity for skew Schur functions*, preprint, September 11, 2026, DOI: 10.5281/zenodo.22723463.